

MAURICE BARAZA OTIENO

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EDUCATION

Lafayette College

Easton, PA

Bachelor of Science Electrical and Computer Engineering

Graduation: May 2026

Relevant Coursework: Power Electronics, Mixed-Signal Design of Electronic Circuits, Microelectronics, Signals and Communication Systems, Digital Circuits and Computer Architecture, Embedded Systems, Industrial Electronics and Control Systems, Electromagnetics, Data Structures and Algorithms, Artificial Intelligence for Computer Vision, Speech and Image Processing

Relevant Skills:

Power converters eg DC-DC converter topologies, PCB design with KiCad and Cadence ORCADX, Signal integrity principles, CMOS Integrated Circuit design principles, BJTs, Filters, OP AMPs, oscillators, circuit simulation (LTSpice and Ngspice), Bench testing with oscilloscopes, DMMs and logic analyzers, soldering & rework, Schematic capture Register Transfer Level Design with Xilinx FPGAs (SystemVerilog synthesis, simulation, and Verification with VIVADO suite), Python scripting, Linux Embedded design with FreeRTOS and bare-metal C/C++, Communication protocols and peripherals: UART, I²C, SPI, ADC, DACs, MATLAB, Visual Studio, git

Awards and Honors: Murray G. Clay '30 Award Fund in Engineering Science, Dean's List: 2 semesters

Campus Engagement: IEEE, Lafayette African and Caribbean Students Association, Soccer Club, and National Society of Black Engineers.

RELEVANT EXPERIENCE

Kenya Power and Lighting Company: Junior Intern, Migori, Kenya

06/2025-08/2025

- Participated in monitoring and inspection of substation components, e.g UPS, Capacitor banks, switch room, to ensure system reliability and safety compliance.
- Actively involved in replacing faulty components of electricity distribution transformers under supervision.

Selected Project Experience *[All projects and personal information will be found at <https://mauriceotieno.com>]*

Smart Technology Rollator for Independence Detection and Empowerment (STRIDE) -Autonomous Stair Climbing system 08/2025 - Present

- Designed the power management system incorporating a 24V battery pack, current monitoring, and protection circuitry for safe motor operation.
- Designed the low-level embedded control architecture integrating STM32 microcontrollers, BLDC motor drivers, and sensor feedback for closed-loop mobility control using the CMSIS V2 FreeRTOS in STM32CUBEIDE.
- Collaborated with seven teammates to perform hardware integration, PCB design, and system-level debugging across analog, digital, and power subsystems.

Pulse Width Modulation Signal generating Voltage Controlled oscillator

01/2025– 3/2025

- Designed a PWM-generating VCO based on a charge/discharge cycle of a capacitor using analog IC op amps, matched CMOS transistors as current mirrors, comparators, and RS NAND latches.
- Simulated the design's performance in LTSPICE and implemented the design on a printed circuit board via a schematic-driven layout in KICAD.
- Performed hardware bring-up and debug using an oscilloscope and function generator, correlating measured performance with simulation

https://github.com/otienomaurice/VOLTAGE_CONTROLLED_OSCILLATOR.git

Pulse Monitor and Reaction Timer

10/2022-12/2022

- Collaborated with a partner to Design and implement a SystemVerilog hardware module for pulse detection and reaction-time measurement on a Nexys A7100T FPGA.
- Verified functional correctness and timing behavior through waveform simulation and testbench development in Vivado
- Integrated and debugged the design on hardware via PMOD interfaces, resolving timing and synchronization issues.

https://github.com/otienomaurice/Pulse_Sensor_and_reaction_timer-.git

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Equity Group Foundation: Intern, College Counselling

05/2022-08/2022

- Mentored recent high school graduates on academic planning, university applications, and career development.
- Provided structured guidance sessions focused on goal setting, scholarship preparation, and transition to tertiary education.

ADDITIONAL EXPERIENCE

ECE Research Fellow, Lafayette College

06/ 2025-06/2026

Student IT Assistant, Tech Lounge, Lafayette College

01/ 2024-06/2026

Languages: English (Fluent), Swahili (Fluent), Luo (Fluent), Sheng (Conversational)